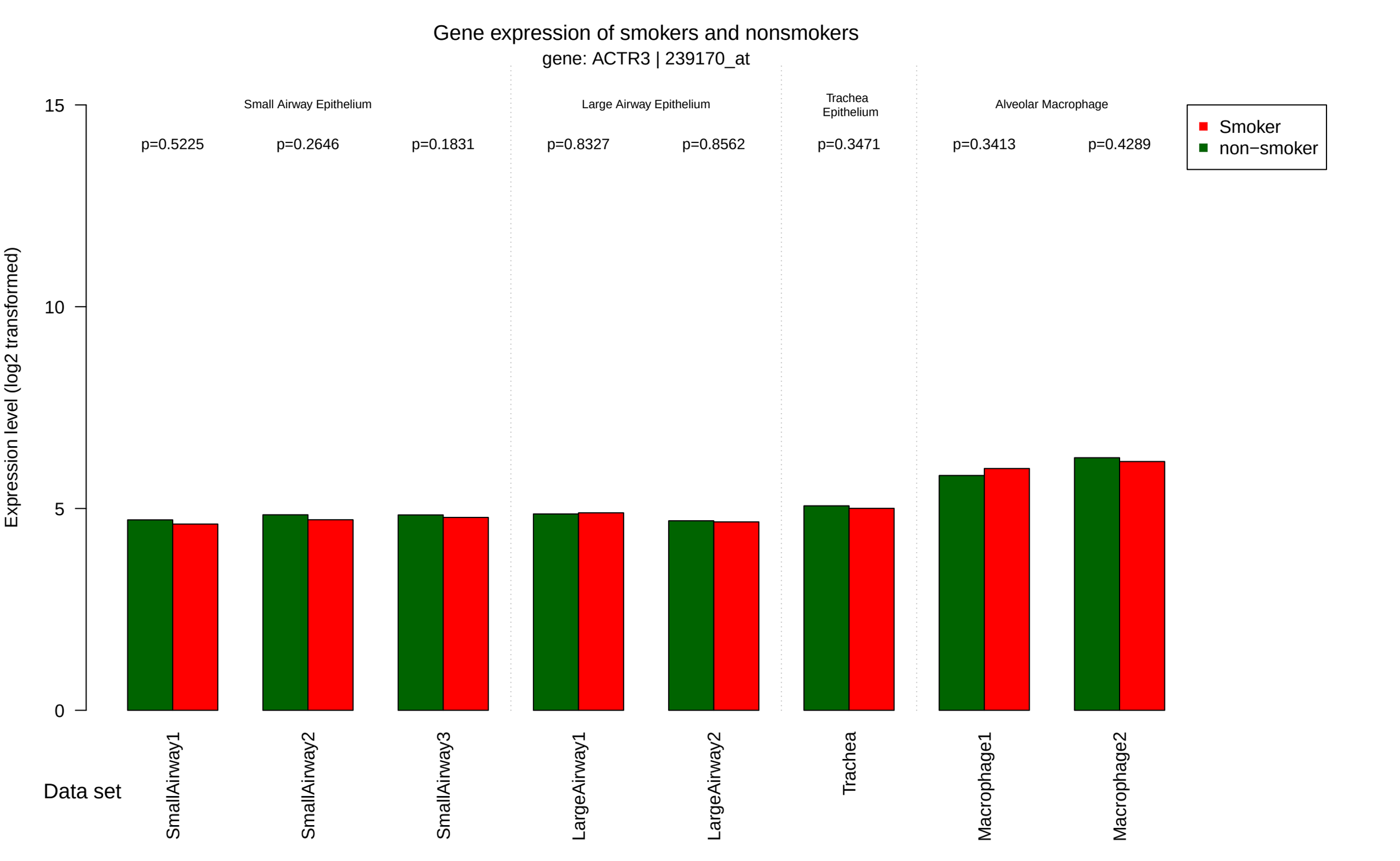
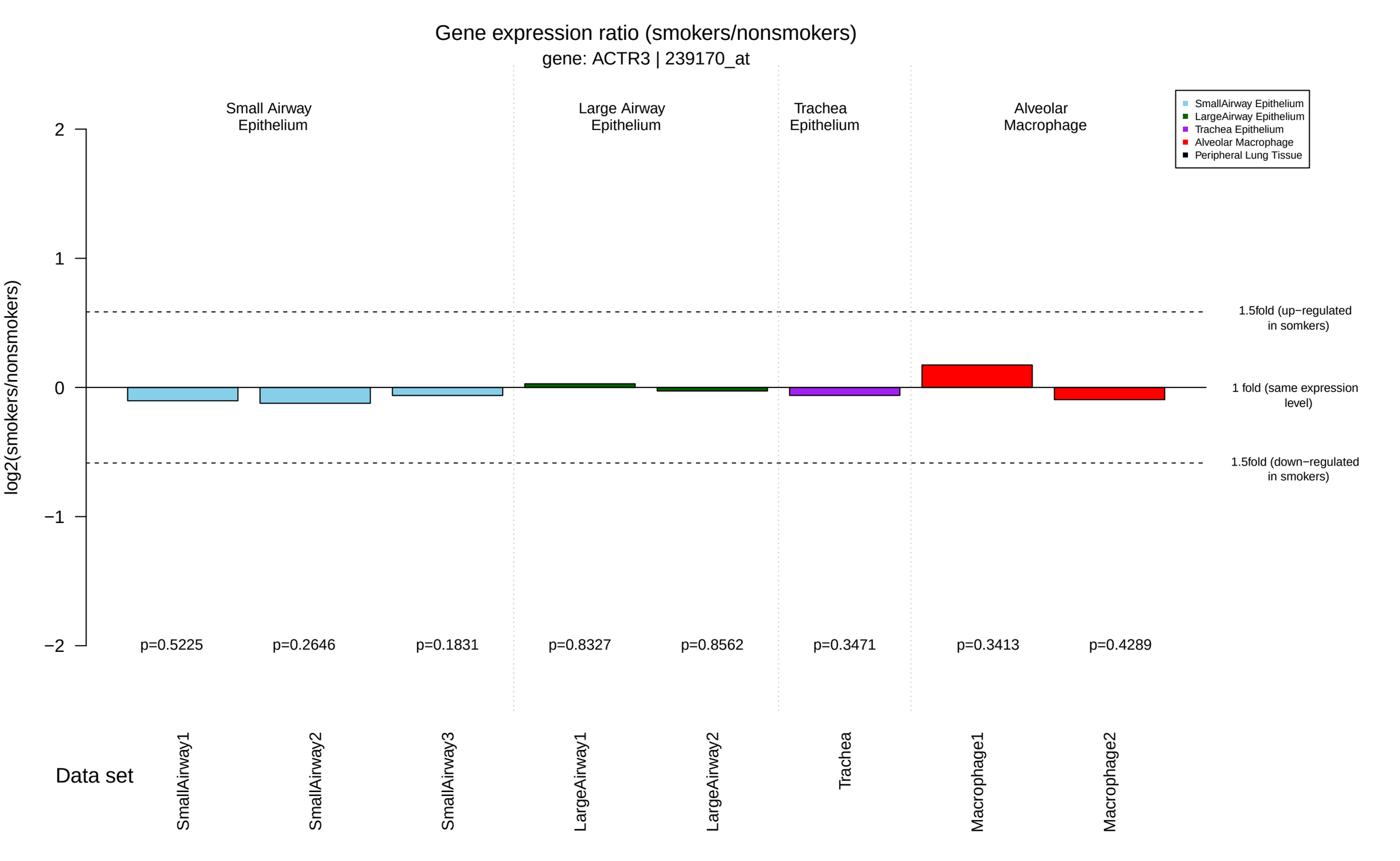
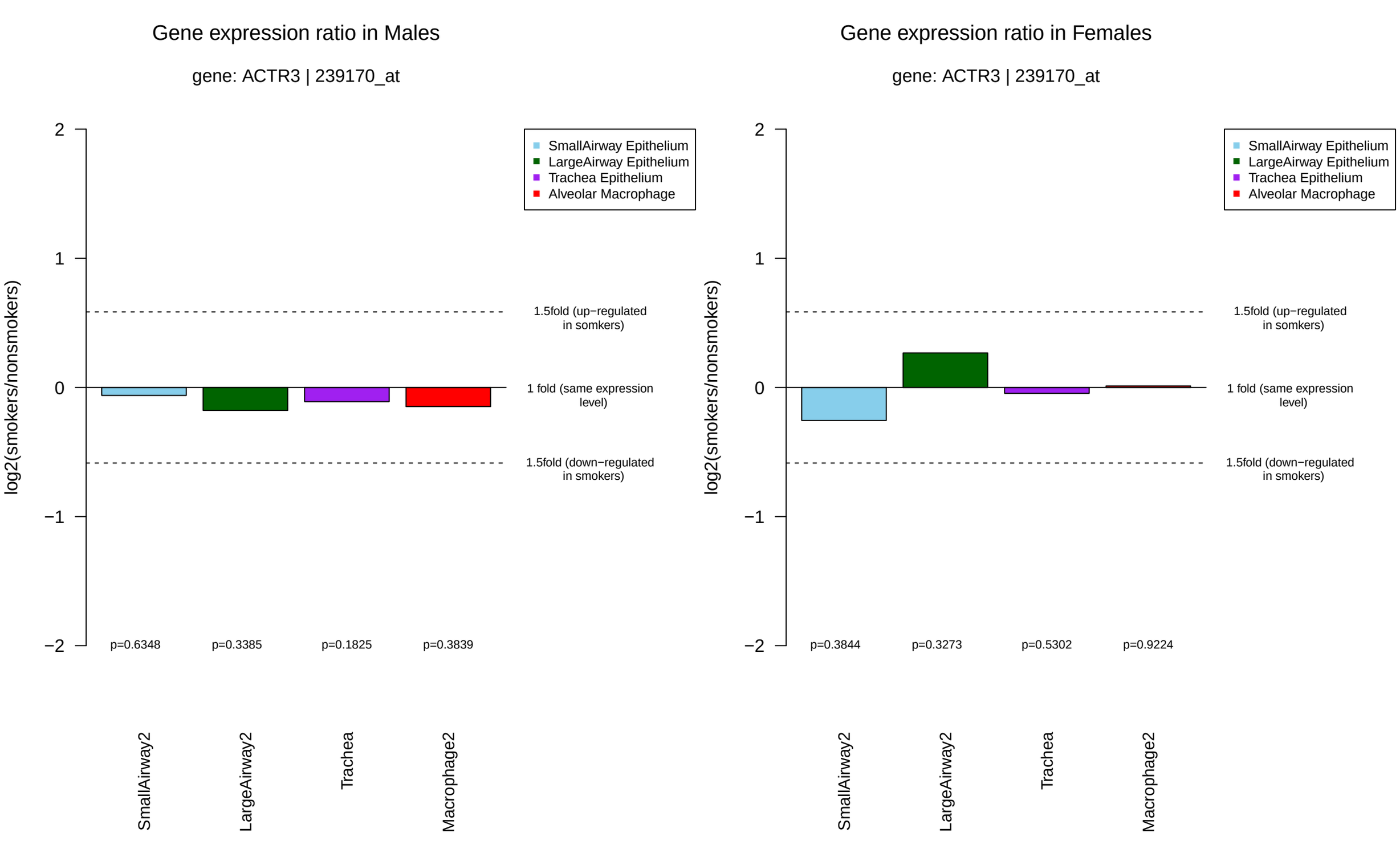
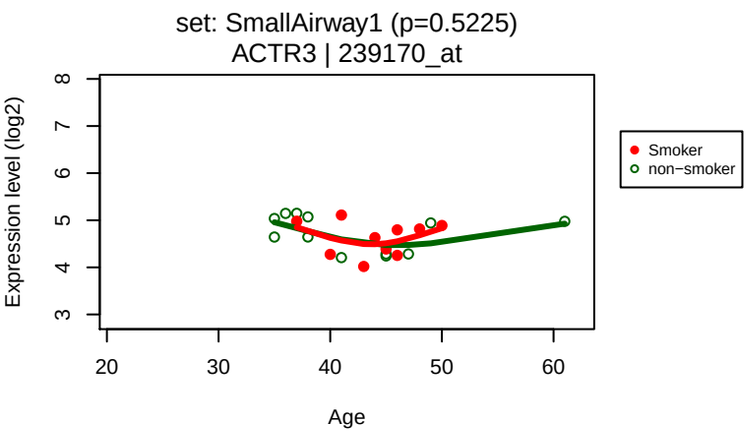
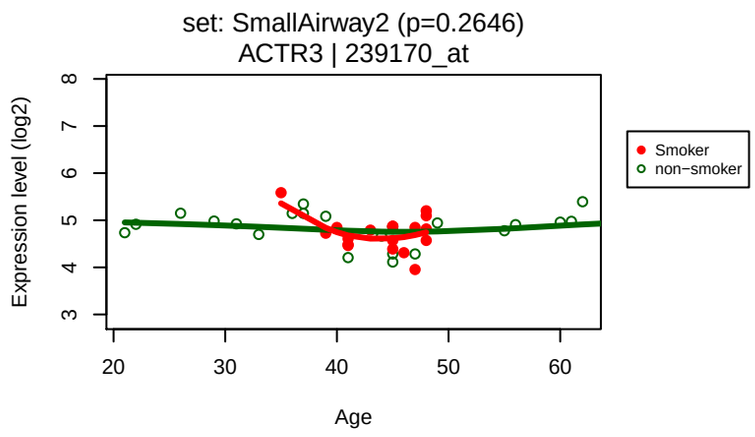
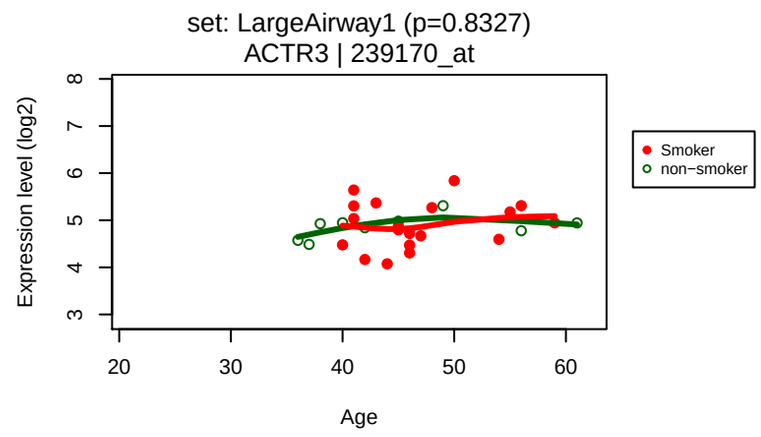

Gene expression of smokers and nonsmokers
gene: ACTR3 | 239170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5225
p=0.2646
p=0.1831
p=0.8327
p=0.8562
p=0.3471
p=0.3413
p=0.4289
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

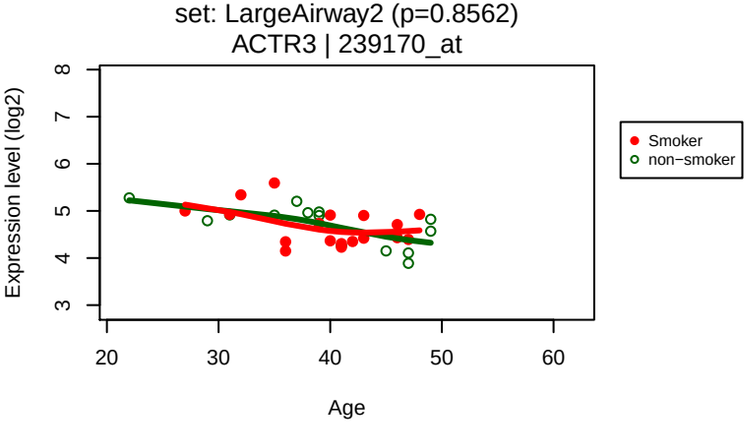

Gene expression ratio (smokers/nonsmokers)
gene: ACTR3 | 239170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5225
p=0.2646
p=0.1831
p=0.8327
p=0.8562
p=0.3471
p=0.3413
p=0.4289
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

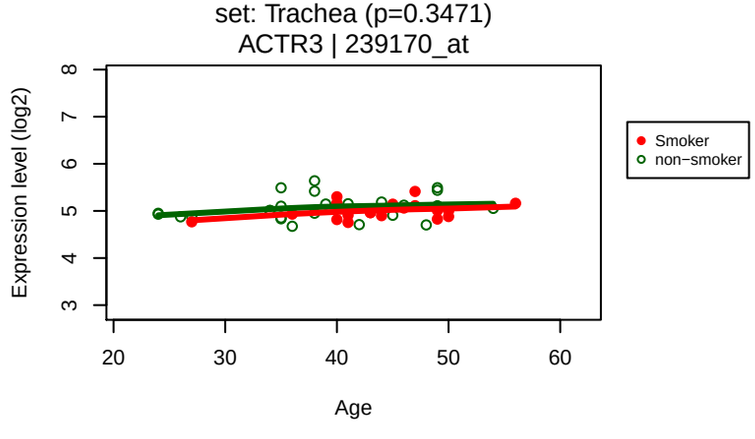

Gene expression ratio in Males
gene: ACTR3 | 239170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6348
p=0.3385
p=0.1825
p=0.3839
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTR3 | 239170_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3844
p=0.3273
p=0.5302
p=0.9224
SmallAirway2
LargeAirway2
Trachea
Macrophage2

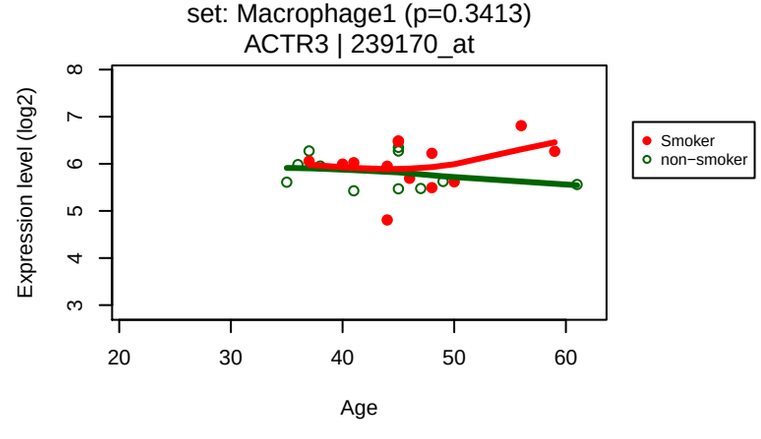

set: SmallAirway1 (p=0.5225)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker

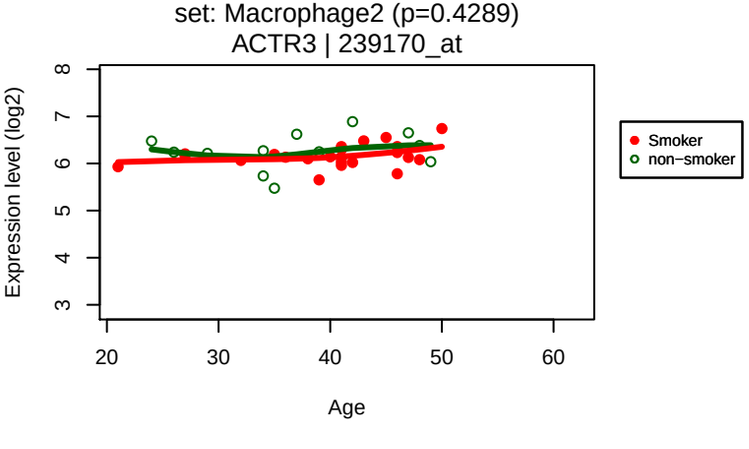

set: SmallAirway2 (p=0.2646)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker

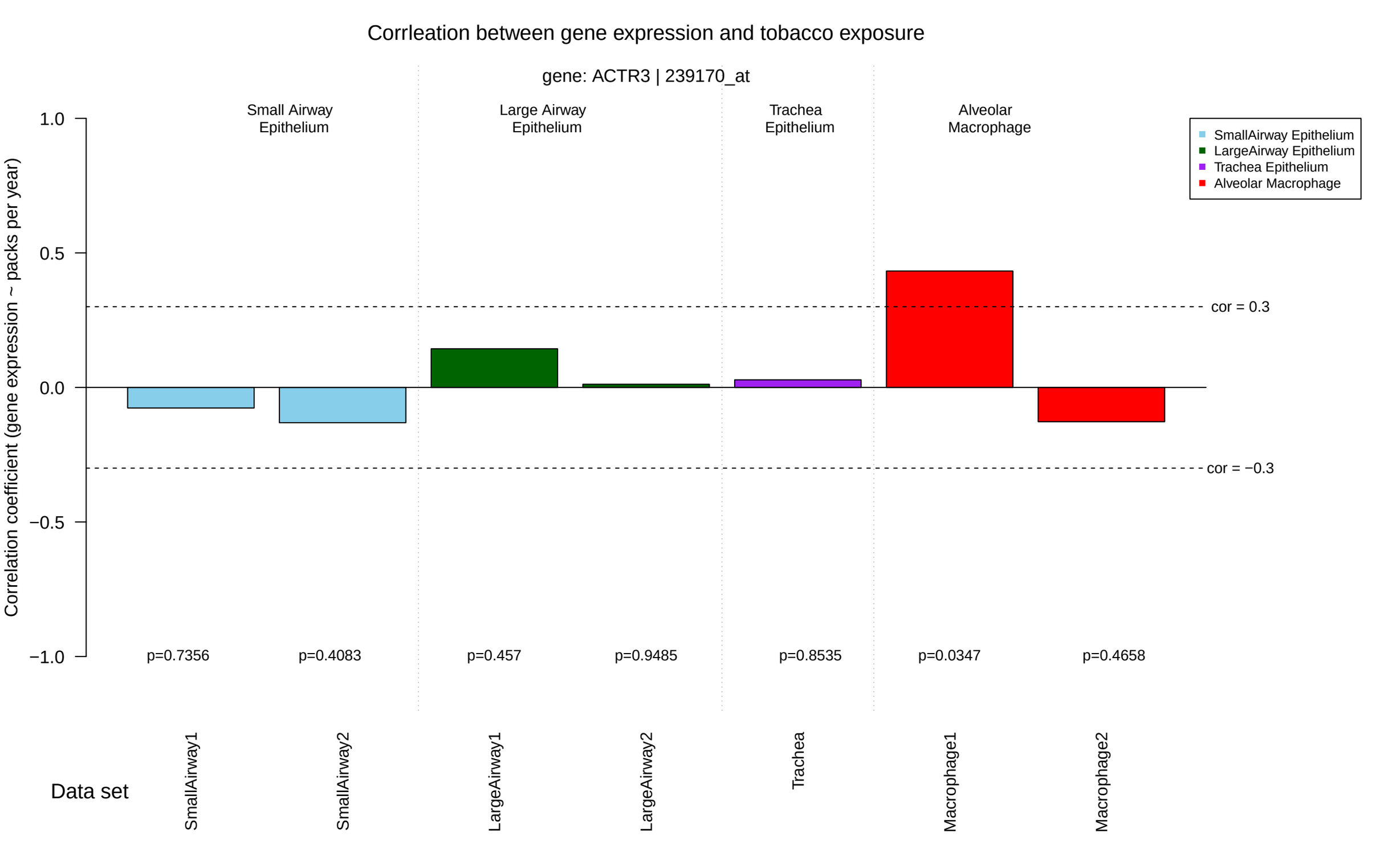

set: LargeAirway1 (p=0.8327)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8562)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3471)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3413)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4289)
ACTR3 | 239170_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTR3 | 239170_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7356
p=0.4083
p=0.457
p=0.9485
p=0.8535
p=0.0347
p=0.4658
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2